

Calibration of reaction-outcome probabilities

Calibration used 2,595 held-out reactions (1,320 positive; 1,275 negative) from 1,091 publications. Checkpoint selection used separate validation data. Fine-tuned open weight LLM denotes MOFinder (step 4500); Base open weight LLM denotes gpt-oss-20b; and Base proprietary LLM denotes GPT-4.1 (gpt-4.1-2025-04-14).

For reaction i , p_i is the positive probability from normalized exact P/N scores at the output position; y_i equals 1 for P and 0 for N. Each z score is a raw logit or token log-probability. Normalization defines a conditional binary probability, whose calibration must be assessed against observed labels:

$$p_i = \frac{\exp(z_{i,P})}{\exp(z_{i,P}) + \exp(z_{i,N})}$$

$$\text{ECE}_M = \sum_{m=1}^M \frac{n_m}{n} |\bar{p}_m - \bar{y}_m|$$

$$\text{Brier} = \frac{1}{n} \sum_{i=1}^n (p_i - y_i)^2$$

For M equal-width bins on $[0,1]$, n_m is the count in bin m ; $\bar{p}_m$ and $\bar{y}_m$ denote mean probability and positive fraction. ECE measures their record-weighted absolute discrepancy; the Brier score measures overall probabilistic error. Table Sx reports $M = 10$; Figure Sx examines binning sensitivity. No post-hoc calibrator was fitted.

Intervals use 5,000 paired DOI-cluster bootstrap replicates (seed 20261002), retaining all reactions within each sampled DOI. These intervals condition on the saved predictions and exclude API rerun variability. GPT-4.1 used temperature 0, top_p 1, a two-token limit, seed 7, and 20 returned log-probability alternatives. Unreturned label tails were bounded without changing the reported digits. Comparisons use direct-label scoring; local base gpt-oss-20b uses GGUF MXFP4 without generated reasoning, whereas archived MOFinder inference precision is unrecorded. Results describe these implementations and curated labels, including reconstructed negatives. Publication overlap across training and evaluation limits generalization to unseen publications and prospective experiments.

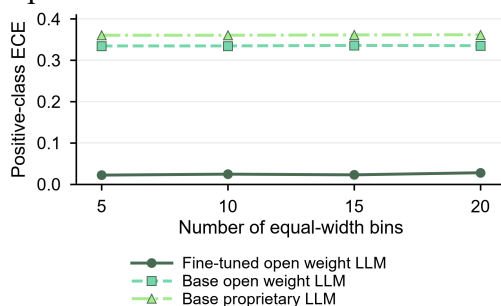


Figure Sx. Positive-class ECE across fixed equal-width binning schemes. All models use the same 2,595 reactions.

Table Sx. Calibration metrics with 95% DOI-cluster bootstrap intervals.

Model	Positive-class ECE ₁₀	Brier score
Fine-tuned open weight LLM	0.0243 (0.0196–0.0466)	0.1209 (0.1082–0.1342)
Base open weight LLM	0.3347 (0.2954–0.3771)	0.3657 (0.3337–0.3979)
Base proprietary LLM	0.3605 (0.3224–0.4040)	0.3802 (0.3428–0.4195)

[Code and prediction CSV: github.com/zzhenglab/MOFinder](https://github.com/zzhenglab/MOFinder)